# Econometrics Notes VI: Elementary Statistics - Part 3: Distributions - Exponential, Gamma, $\chi^2$

Artur Wegrzyn

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**Streszczenie**

This note presents basic results concerning distributions that will show up frequently in later considerations: exponential, gamma and $\chi^2$.

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## 1 Characteristic Function Uniquely Determines Probability Distribution

In this note the fact that characteristic function uniquely determines probability distribution is used. For proof and comprehensive discus-

sion of this theorem cf.:

- in English: [1] - chapter XV: *Characteristic Functions*
- in Polish: [2] - chapter 9: *Funkcje charakterystyczne*

In these notes, however, I will work with moment-generating functions instead of characteristic functions for the sake of clarity.

# 2 Exponential Distribution

Random variable $X$ follows exponential distribution with rate[1] parameter $\lambda$ if its pdf is:

$$f_X(x) = \lambda e^{-\lambda x}$$

MGF of $X$ is:

$$M_X(t) = E(e^{tX}) =$$

The fact that $X$ is exponentially distributed with scale parameter lambda is denoted:

$$X \sim Exp(1/\lambda)$$

# 3 Gamma Distribution

## 3.1 Sum of $p$ Exponential Random Variables

Let $X_1, X_2, \ldots, Y_p$ be a $p$-element sample drawn indepentently from an exponential distribution with expected value $1/\lambda$ i.e.:

$$X_i \sim Exp(1/\lambda)$$

## 3.2 •

# 4 Square of Standard Normal Distribution

Suppose that $U$ follows standard normal distribution:

$$U \sim N(0, 1)$$

[1] Rate parameter is inverse of scale parameter. Recall that $s$ is a scale parameter if $F(x; s, \theta) = F(x/s; 1, \theta)$

Consider $Z = U^2$. Then:

$$P(Z \leq z) = P(U^2 \leq z) = P(-\sqrt{z} < U \leq \sqrt{z})$$

Hence:

$$f_Z(z) = \frac{dF_Z(z)}{dz} = \frac{d}{dz} P(-\sqrt{z} < U \leq \sqrt{z})$$

A little bit of algebra yields:

$$f_Z(z) =$$

# 5 $\chi^2$ Distribution

In this and subsequent section we consider a sample $X_1, X_2, \ldots, X_n$ that is drawn independently from a normal distribution with mean $\mu$ and variance $\sigma^2$:

$$X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$$

# Literatura


[1] W. Feller, *An Introduction to Probability Theory And Its Applications, Volume II, 2nd Edition.* Wiley, 1968.

[2] S. R. Jakubowski J., *Wstęp do teorii prawdopodobieństwa.* SCRIPT, 2010, Warszawa.